

Probability Exam

Questions are taken from past papers of STEP (entry test to Cambridge), Section C: Probability and Statistics. You will get bonus points for problems completely solved.

Solving the following problem is obligatory. Solve this problem first please as completely as you can.

STEP 2000, Paper II, Question 12

Tabulated values of $\Phi(\cdot)$, the cumulative distribution function of a standard normal variable, should not be used in this question.

Henry the commuter lives in Cambridge and his working day starts at his office in London at 09:00. He catches the 07:15 train to King's Cross with probability p , or the 07:20 to Liverpool Street with probability $1 - p$. Measured in minutes, journey times for the first train are $N(55, 25)$ and for the second are $N(65, 16)$. Journey times from King's Cross and Liverpool Street to his office are $N(30, 144)$ and $N(25, 9)$, respectively. Show that Henry is more likely to be late for work if he catches the first train.

Henry makes M journeys, where M is large. Writing A for $1 - \Phi(20/13)$ and B for $1 - \Phi(2)$, find, in terms of A , B , M and p , the expected number, L , of times that Henry will be late and show that for all possible values of p ,

$$BM \leq L \leq AM.$$

Henry noted that in $3/5$ of the occasions when he was late, he had caught the King's Cross train. Obtain an estimate of p in terms of A and B .

[A random variable is said to be $N(\mu, \sigma^2)$ if it has a normal distribution with mean μ and variance σ^2 .]

Solving the following two problems are optional – don't worry if you can't solve them both because your time is restricted.

STEP 1998, Paper II, Question 13

A random variable X has the probability density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Show that¹

$$P(X > s + t \mid X > t) = P(X > s).$$

The time it takes an assistant to serve a customer in a certain shop is a random variable with the above distribution and the times for different customers are independent. If, when I enter the shop, the only two assistants are serving

¹That's why we can say that the exponential distribution is "memoryless" or "forever young". It means the distribution of a "waiting time" until a certain event does not depend on how much time has elapsed already.

one customer each, what is the probability that these customers are both still being served at time t after I arrive?

One of the assistants finishes serving his customer and immediately starts serving me. What is the probability that I am still being served when the other customer has finished being served?

STEP 1998, Paper II, Question 14

The staff of Catastrophe College are paid a salary of A pounds per year. With a Teaching Assessment Exercise impending it is decided to try to lower the student failure rate by offering each lecturer an alternative salary of $B/(1 + X)$ pounds, where X is the number of his or her students who fail the end of year examination. Dr Doom has N students, each with independent probability p of failure. Show that she should accept the new salary scheme if

$$A(N + 1)p < B(1 - (1 - p)^{N+1}).$$

Under what circumstances² could X , for Dr Doom, be modelled by a Poisson random variable? What would Dr Doom's expected salary be under this model?

²Don't spend more than a minute with this theoretical question — go on using Poisson distribution as a model and writing down Dr Doom's salary! Later, when you have time, you can check exact and approximate salary of Dr Doom in different cases (e.g. in an Excel sheet), how close they are.